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Prevalence of illness, poor mental health and sleep quality and low energy availability prior to the 2016 Summer Olympic Games

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ABSTRACT

Objective Establish the prevalence of illness symptoms, poor sleep quality, poor mental health symptoms, low energy availability and stress-recovery state in an Olympic cohort late in the 3 months prior to the Summer Olympic Games.

Methods Olympic athletes (n=317) from 11 sports were invited to complete questionnaires administered 3 months before the Rio 2016 Olympic Games. These questionnaires included the Depression, Anxiety and Stress Questionnaire, Perceived Stress Scale, Dispositional Resilience Scale, Recovery-Stress Questionnaire (REST-Q-52 item), Low Energy Availability in Females Questionnaire (LEAF-Q), Epworth Sleepiness Scale, Pittsburgh Sleep Quality Index and custom-made questionnaires on probiotic usage and travel. Multiple illness (case) definitions were applied. ORs and attributable fractions in the population were used. Factor analyses were used to explore the relationships between variables.

Results The response rate was of 42% (male, n=47, age 25.8±4.1 years; female, n=85, age 24.3±3.9 years). Low energy availability was associated with sustaining an illness in the previous month (upper respiratory, OR=3.8, 95% CI 1.2 to 12). The main factor relating to illness pertained to a combination of anxiety and stress-recovery states (as measured by the REST-Q-52 item). All participants reported at least one episode of illness in the last month (100% prevalence).

Conclusions All participants reported at least one illness symptom in the previous month. Low energy availability was a leading variable associated with illness in Olympic-class athletes. The estimates duration of symptoms ranged from 2 to 7 days. Factor analyses show the interdependence of various health domains and support multidisciplinary care.

INTRODUCTION

Health is complex and forms part of an overarching system[1] that effects other aspects of an integrated performance model[2] influencing outcomes of success and failure.[3 4] During the competition period, the incidence of illness is typically between 5% and 17%.[5 6] Respiratory illness can be responsible for up to 50% of cases.[5] Many athletes experience illness in the preparations leading to competition[7 8] resulting in modification of training that in turn could impair performance outcomes.[7]

Given the need for athletes to remain healthy to complete the training required for success, a periodical health examination was implemented in December 2015–January 2016 (9 months before the Olympic Games) in Australian Olympic shadow squad members.[9] At this time point in the Olympic preparations, low energy availability (LEA), mental health and female sex were associated with high attributable risks of suffering a sport-incapacity illness in the previous month.[9] However, this work was undertaken in a small sample (n=81) early in the preparation phase, and replication in a larger cohort is warranted to provide attention in greater detail to the type of illness and underpinning factors. This is the first study to investigate multifactorial risk factors, and prevalence of aspects of health, in an Olympic population 3 months prior to the Olympic Games.

This study will inform practitioners in four ways:

1. establish the prevalence of illness symptoms, independent of medical attention and sports incapacity in an Olympic cohort late in the preparations (3 months prior) for the Summer Olympic Games
2. establish the prevalence of poor sleep quality, poor mental health symptoms, LEA and stress-recovery state in an Olympic cohort late in the preparations (3 months prior) for the Summer Olympic Games
3. describe the association of these factors to a recent history of injury and illness (within the last month)
4. provide real-world case studies of how these factors coexist supported by factor analyses.

METHODS

We employed a cross-sectional study, analysed as a case–control design. Questionnaires were administered, as previously described,[9] via an electronic management system used for routine collection of internal training loads and wellness monitoring in Australian Olympic athletes. All questionnaires were completed during a 1-month period (20 April 2016–31 May 2016) approximately 3 months before the 2016 Rio Olympic Games. The date of administration for the questionnaires was determined by relevant staff from the Sporting Organisation, with athletes being able to fill them out at any point during the survey period. Athletes were

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Table 1 Summary of the questionnaires administered during preparations for the Olympic Games

Domain	Questionnaire	Measure
Probiotics	Custom questionnaire	Probiotic use (any brand)
Travel	Custom questionnaire	Number of travel occasions
Psychology	Depression, Anxiety, Stress Scale	Symptoms of depression, anxiety and stress.[25 26]
	Dispositional Resilience Scale	Resilience to stress.[27]
	Recovery-Stress Questionnaire[28]	Multidimensional analysis of an individual's stress-recovery state. For analysis, the subscales scores for overall and sport-specific stress and recovery were analysed independently.
	Perceived Stress Scale	Level of global perceived psychological stress.[29]
Sleep	Pittsburgh Sleep Quality Index[30]	Sleep quality, latency, duration, efficiency and disturbances. For analysis, a cut-off score of ≥5 was used.
	Epworth Sleepiness Scale[31]	Daytime sleepiness symptoms. For analysis, a cut-off score of ≥10 was used.
Nutrition	Low Energy Availability in Females[15]	Injury history, gastrointestinal and reproductive function. For analysis, a global score ≥8 was used to indicate increased risk of low energy availability[15] as well as the individual components: gastrointestinal: ≥4, injury: ≥4 and reproductive function: ≥8.

See Drew *et al*[8] for details.

recruited through their National Sporting Organisation, which provided organisational consent. This study complies with the Declaration of Helsinki, with athletes being recruited through their National Sporting Organisation, which provided organisational consent.

Definition of an illness and injury

Multiple definitions aligned to the Injury Definitions Concept Framework were employed and analysed in this study.[10] Sports incapacity was defined as the self-report of missed training for a period of 24 or more consecutive hours due to medical illness or injury in the month prior to completion of the questionnaires. An athlete self-report without sports incapacity was defined as reporting of signs or symptoms in the previous month that did not involve missed or modified training for greater than 24 hours. A control was defined as a participant sampled from the same cohort who self-reported completion of all training days uninterrupted by illness (types) in the month prior to the study. These participants are described as 'remaining full training'.

Questionnaires

Eight questionnaires were administered, as shown in table 1. These were categorised as psychology, sleep, travel, probiotic use or nutrition. Prospectively collected internal training loads using the s-Rate of Perceived Effort method[11] were also retrieved from the online monitoring system; however, these were not analysed due to incomplete records.

Statistics and data handling

Prior to descriptive and inferential analyses, data were assessed for normality using the Shapiro-Wilk test statistic and visual inspection. Results are presented as mean and SD for parametric data, and median and IQR for non-parametric data. All statistical analyses were undertaken in Stata V.13. The statistical significance level was set a priori where 95% CIs did not include 1.0. All estimates are reported in relation to their clinical importance. All analyses were completed using a custom Stata script such that the results can be reproduced.

Estimates of prevalence and risks

Prevalence was calculated as the number of responses above the cut-off value divided by the total number of responses. ORs with associated attributable fractions in the exposed and attributable fractions in the population (AFP) with 95% CIs (using Fisher's exact) were calculated using separate two-by-two tables for binary data and continuous data with published thresholds (see table 1). The attributable fraction (percentage) that is the proportion of disease attributable to the marker and/or to factors associated with it.[12]

Factor and principal component analyses

Exploratory factor analysis (FA) was then performed to assess inter-relationships between measures of Depression, Anxiety and Stress Questionnaire (DASS-21) (depression, anxiety and stress), dispositional resilience (hardiness), Low Energy Availability in Females Questionnaire (LEAF-Q), Recovery-Stress Questionnaire (REST-Q) overall stress, REST-Q overall recovery, REST-Q sport-specific stress, REST-Q sport-specific recovery, Epworth Sleepiness Scale (sleepiness) and Pittsburgh Sleep Quality Index. All data were analysed as continuous variables. As this sample size was greater than 50, exploratory FA was deemed to be appropriately justified.[13] The number of factors was limited to three (which had eigenvalues greater than 1.0), and an item-loading threshold greater than or equal to 0.5 was considered adequate for inclusion in a factor.

RESULTS

Elite athletes (n=206) consented to participate from an estimated eligible population of 317, representing a response rate of 65%. However, only 132 completed the protocol with sufficient detail for analysis of the health outcomes (male, n=47, age 25.8±4.1 years; female, n=85, age 24.3±3.9 years; mean±SD). Responses (n) to the individual questionnaires are presented in table 2. The distribution of respondents by sport was: boxing: n=1; equestrian: n=1; football (soccer): n=20; gymnastics: n=1; hockey: n=43; rowing: n=13; rugby sevens: n=32; sailing: n=1; triathlon: n=2; water polo: n=15; unreported: n=3. No athletes reported smoking cigarettes.

Prevalence of conditions and abnormal results

The point prevalence of poor mental health was 14%–17%, and poor sleep quality was 49% in this cohort. The prevalence of LEA was also high in females with 40% scoring above the clinical cut-off value (≥8). The prevalence of health domains are presented in table 2. The prevalence of symptoms and sports incapacity is reported in table 3 and figure 1.

The association between health domains and illness

We investigated a priori the associations between any illness and the factors in table 1. Notably, all athletes in this study reported symptoms within the previous month. The associations between explanatory factors and athlete self-reported illness categories are presented in table 4. Relating to athlete self-reported illness (independent of sports incapacity), LEA was associated with higher odds of reporting symptoms of upper respiratory tract infection (OR=3.8, 95% CI 1.3 to 12), bodily aches (OR=5.8,

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Table 2 Prevalence of aspects of health or potential determinants of health

Domain	Questionnaire	Responses (n) Yes	Responses (n) No	Prevalence (%)
Probiotics	Any usage	38	95	29
Travel in the last month	>1 occasion	57	76	43
	>2 occasions	16	117	12
	>3 occasions	5	128	3.8
Psychology	Depression (DASS-21 >4)	22	105	17
	Anxiety (DASS-21 >3)	24	103	19
	Stress (DASS-21 >8)	18	109	14
	Dispositional Resilience Scale (DRS <27)	13	115	10
	Perceived Stress Scale			
	Recovery* (REST-Q)[28]	–	–	–
Sleep	Pittsburgh Sleep Quality Index[30]	51	53	49
	Epworth Sleepiness Scale[31] (>10)	23	84	22
Nutrition	Low Energy Availability in Females Questionnaire (≥8)[15]	36	54	40

*Cut-off values not available.
DASS-21, Depression, Anxiety and Stress Questionnaire; REST-Q, Recovery-Stress Questionnaire.

95% CI 1.6 to 24), gastrointestinal disturbances (OR=3.8, 95% CI 1.3 to 12) and head symptoms (OR=4.4, 95% CI 1.4 to 124). Poor sleep quality (Pittsburgh Sleep Quality Index (PSQI) ≥5) was associated with higher odds of having gastrointestinal disturbance. All other explanatory factors were not significantly related to any other illness category.

Factor analyses

Factors independent of health status

Two significant factors were identified (factor 1: eigenvalue=2.24, proportion=0.56; factor 2: eigenvalue=1.61, proportion=0.39) with a cumulative proportion of 0.96 (table 5). Factor 1 pertains depression, anxiety and stress (as assessed by the DASS-21), whereas factor 2 pertains primarily to dispositional resilience and overall recovery and overall stress (as assessed by the REST-Q). These two factors combined explain 96% of variance in the data. The factor loadings are presented in table 5.

Factors relating to recent sports incapacity illness

Two significant factors were identified in relation to illness (factor 1: eigenvalue=2.82, proportion=0.45; factor 2: eigenvalue=1.84, proportion=0.29) with a cumulative proportion of 0.74. Factor 1 pertained to anxiety (DASS-21) and the overall recovery, overall and stress and sport-stress states measured by the REST-Q), whereas factor two related to depression, stress (DASS-21) and overall recovery as measured by the (REST-Q). These two factors combined explain 74% of the variance in the data.

Factors relating to uninterrupted training

Five significant factors were identified in relation to remaining healthy (factor: eigenvalue, proportion: factor 1: 4.18, 0.32; factor 2: 2.13, 0.42; factor 3: 1.71, 0.13; factor 4: 1.38, 0.11; factor 5: 1.19, 0.09) with a cumulative proportion of 0.82. The leading two factors are presented in table 5. Factor 1 pertained to depression, anxiety and stress (DASS-21), resilience (DRS) and sleep quality (PSQI), whereas factor 2 related to LEA,

Table 3 Monthly prevalence recorded 3 months before the Olympics (April–May 2016)

Symptom category	Definition	Responses Yes	Responses No	Prevalence (%)
Upper respiratory	Any symptoms	61	71	46
	Sports incapacity	58	3	95
	Number of days			
	Full (n=59)	4.4±4.6		97*
	Mod (n=20)	0.9±1.6		33*
	No (n=17)	0.7±1.4		28*
Bodily aches	Any symptoms	26	106	20
	Sports incapacity	26	1	96
	Number of days			
	Full (n=25)	6.5±5.6		96*
	Mod (n=5)	0.5±1.0		19*
	No (n=4)	0.3±0.9		15*
Gastrointestinal	Any symptoms	28	104	21
	Sports incapacity	22	6	79
	Number of days			
	Full (n=22)	4.1±6.4		79*
	Mod (n=6)	0.4±0.8		21*
	No (n=9)	0.7±1.2		32*
Head	Any symptoms	31	101	32
	Sports incapacity	27	4	87
	Number of days			
	Full (n=27)	2.4±2.0		87*
	Mod (n=8)	0.4±0.8		26*
	No (n=7)	0.3±0.7		23*
Chest	Any symptoms	10	122	7.6
	Sports incapacity	10	0	100
	Number of days			
	Full (n=10)	6.8±7.2		100*
	Mod (n=3)	0.5±1.0		30*
	No (n=3)	0.5±0.8		30*
Fatigue	Any symptoms	48	84	36
	Sports incapacity	45	3	94
	Number of days			
	Full (n=45)	7.1±6.3		94*
	Mod (n=8)	0.6±1.5		17*
	No (n=2)	0.2±0.9		42*
Eye	Any symptoms	10	122	7.6
	Sports incapacity	8	2	80
	Number of days			
	Full (n=8)	2.9±2.5		80*
	Mod (n=1)	0.1±0.3		10*
	No (n=0)	0±0		0.0*
Other illness	Any symptoms	2	132	1.5
	Sports incapacity	2	0	100
	Number of days			
	Full (n=1)	0.5±0.7		50*
	Mod (n=2)	3.0±2.8		100*
	No (n=1)	1.5±2.1		50*
Injury	Number of injuries			
	(Range)	0–3		
	(Mean±SD)	0.4±0.6		
	Sports incapacity	31	101	24
	Medical attention	57	75	43
	Medical treatments	1.9±3.5		

*Note this is the prevalence among those with symptoms.
Number of days with symptoms expressed as mean±SD. Type of training: Mod, modified training due to symptoms; No, no training due to symptoms; Full, full training with symptoms.

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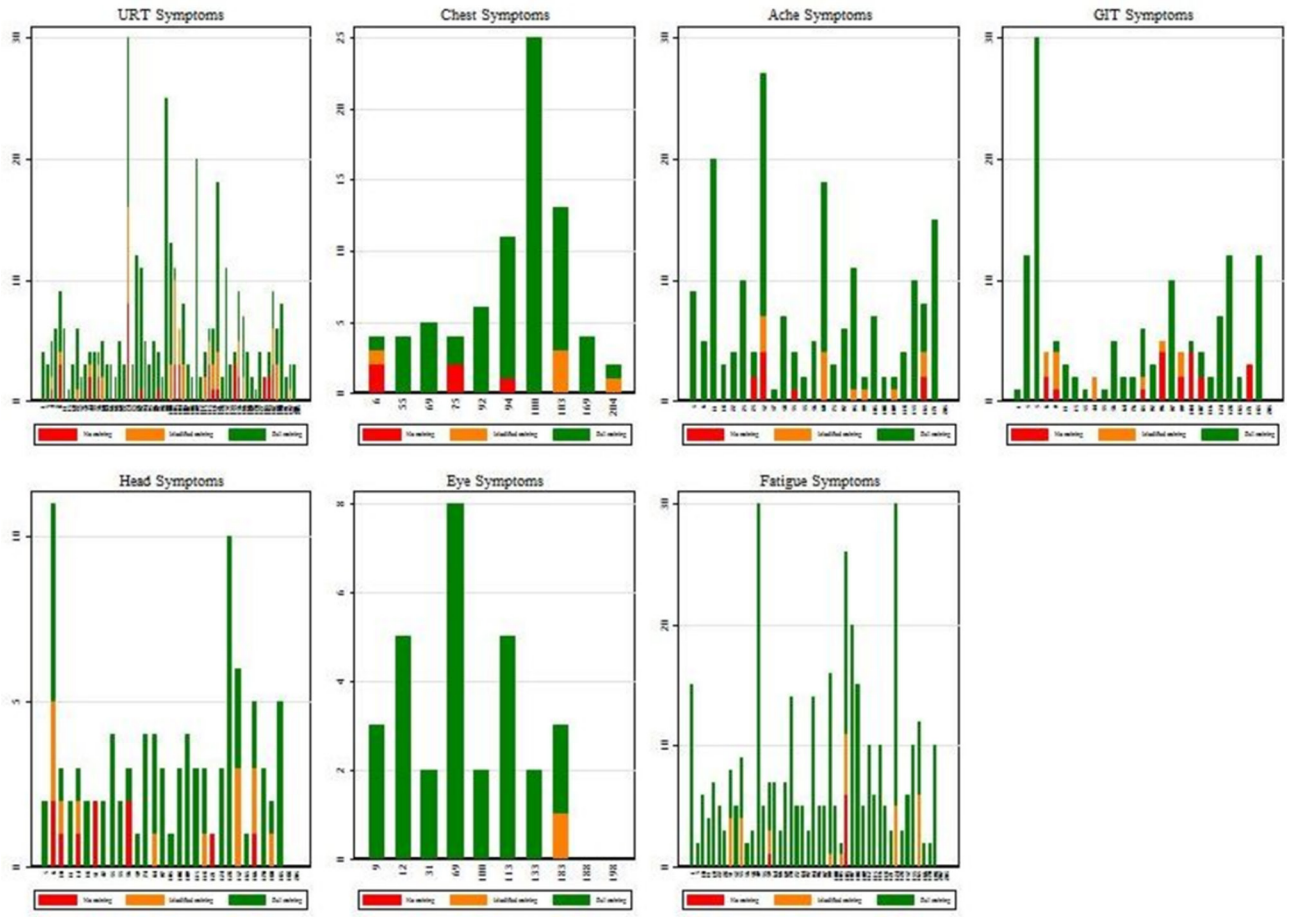

Figure 1 Distribution of self-reported illnesses.

overall stress (REST-Q) and the number of travels in the previous month. These two factors combined explain 49% of the variance in the data.

DISCUSSION

This study highlights a high point prevalence of athlete self-reported illnesses in the preparations (3 months prior) leading to an Olympic Games, with 100% of participants reporting at least one illness symptom in the previous month. LEA (as assessed by the LEAF-Q) is a leading variable associated with illness in Olympic-class athletes in accordance with previous observations in this cohort.[9] New data from the current study provide useful estimates of the duration of symptoms (~2–7 days) and modified training related to the types of illness experienced by Olympic-level athletes. While limited to self-reports, the patterns of illness and underlying associations with sleep, energy availability and mental health provide clinicians with key measures to inform return-to-sport decisions and importantly prevention programmes prior to any incidence. The results of the factor analyses show the interdependence of various health domains and support the proposition that athlete health is best managed by a multidisciplinary team of medical and other practitioners.

LEA as a potential risk factor for illness development

The difficulty of predicting health outcomes from screening data is well established.[14] This study has replicated our earlier study that identified LEA some 9 months prior to the Olympic Games as a risk factor for illness.[9] Given its reproducible magnitude of association (OR between 4 and 8) and stability of measurement,[15] this study confirms the utility of this questionnaire on LEA in identifying potential mechanisms and/or aetiologies of illness in female athletes. Future research should investigate this relationship using laboratory measures[15–18] that directly assess the energy availability status of athletes rather than the indirect methodology employed here. We anticipate that a direct measurement of LEA may identify features that could be included in prevention and management programmes and that, once implemented, could likely reduce the incidence of illness in this population. The high prevalence identified in this and previous work together with the lack of evidence indicating that individualised interventions are better than population-level interventions in sport[14] support the notion that targeted intervention should be provided to all female athletes. Future research should compare the effectiveness of such approaches prior to formulation and implementation of recommendations. It should be noted that the LEAF-Q has been validated in a population of endurance athletes and professional dancers.[15] Wider validation, with specific focus on Summer Olympic sports, is required to further support the results of this study.

Poor mental health is prevalent but was not associated with illness

In our earlier investigation of Olympic squad members, fluctuations in markers of poor mental health were associated with sports incapacity.[9] In the present study, this association was not replicated. Several likely explanations for the different patterns of the mental health–illness relationship are possible. First, the

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Table 4 Association between domains of health and a recent history of illness (all definitions) relating to each symptom category

			URT		Chest		Bodily aches		GIT		Head		Fatigue	
Domain	**Questionnaire**	**Statistic**	**Est.**	**95% CI**	**Est.**	**95% CI**	**Est.**	**95% CI**	**Est.**	**95% CI**	**Est.**	**95% CI**	**Est.**	**95% CI**
Probiotics	Any usage (n=120)	OR	1.9	0.8 to 4.5	1.1	0.2 to 5.0	0.5	0.1 to 1.6	1.9	0.8 to 4.5	1.8	0.7 to 4.6	**3.6**	**1.5 to 8.5**
		AFE	0.5	–0.2 to 0.8	0.06	– 4.9 to 0.8	0.5	–0.6 to 0.9	0.5	– 0.2 to 0.8	0.5	–0.4 to 0.8	0.7	0.3 to 0.9
		AFP	0.17	–	0.02	–	0.1	–	0.2	–	0.17	–	0.33	-
Travel in the last month	>2 occasions (n=120)	OR	0.9	0.3 to 2.9	3.6	0.5 to 18	1.4	0.3 to 5.3	0.9	0.3 to 2.9	1.6	0.4 to 5.5	0.8	0.2 to 2.6
		AFE	0.1	–1.9 to 0.7	0.7	–0.9 to 0.9	0.3	–2.3 to 0.8	0.1	–1.9 to 0.7	0.4	–1.6 to 0.8	0.2	–1.6 to 0.8
		AFP	0.1	–	0.2	–	0.05	–	0.01	–	0.06	–	0.03	–
Psychology	Depression (DASS-21 >4) (n=115)	OR	0.7	0.2 to 2.2	0.5	0.01 to 4.3	0.9	0.2 to 3.2	0.7	0.2 to 2.2	1.3	0.4 to 4.2	1.0	0.3 to 3.0
		AFE	0.3	–1.2 to 0.8	0.5	–3.3 to 1.0	0.1	–2.2 to 0.8	0.3	–1.2 to 0.8	0.2	–1.7 to 0.8	0.03	–2.0 to 0.7
		AFP	0.05	–	0.08	–	0.02	–	0.05	–	0.05	–	<0.01	–
	Anxiety (DASS-21 >3) (n=115)	OR	0.9	0.3 to 2.7	1.1	0.1 to 6.3	0.8	0.2 to 2.7	0.9	0.3 to 2.7	2.5	0.8 to 7.3	1.7	0.6 to 4.9
		AFE	0.05	–1.7 to 0.7	0.1	–8.2 to 0.84	0.2	–1.7 to 0.8	0.05	–1.7 to 0.7	0.6	–0.3 to 0.9	0.4	–0.8 to 0.8
		AFP	0.01	–	0.02	–	0.04	–	0.01	–	0.2	–	0.09	–
	Stress (DASS-21 >8) (n=115)	OR	0.7	0.2 to 2.2	0.6	0.01 to 5.1	1.1	0.2 to 3.9	0.7	0.2 to 2.2	0.5	0.09 to 2.1	0.9	0.3 to 2.9
		AFE	0.3	–1.2 to 0.8	0.4	–4.1 to 1.0	0.06	– 3.4 to 0.7	0.3	–1.2 to 0.8	0.5	–1.1 to 0.9	0.1	–1.9 to 0.7
		AFP	0.05	–	0.06	–	0.01	–	0.05	–	0.08	–	0.02	–
	Dispositional resilience (DRS <27) (n=120)	OR	1.6	0.4 to 6.9	1.0	0.02 to 8.5	0.3	0.01 to 2.3	1.6	0.4 to 6.9	1.0	0.2 to 4.4	0.8	0.2 to 3.3
		AFE	0.4	–1.4 to 0.9	0.0	–47 to 0.8	0.7	–1.3 to 1.0	0.4	–1.4 to 0.9	0.0	–5.2 to 0.8	0.2	–2.3 to 0.8
		AFP	0.05	–	0.0	–	0.08	–	0.05	–	0.0	–	0.02	–
	Perceived Stress Scale (10-item score >20) (n=125)	OR	0.8	0.3 to 2.3	0.5	0.1 to 4.2	0.2	<0.01 to 1.1	0.8	0.3 to 2.3	0.7	0.2 to 2.3	0.7	0.2 to 2.0
		AFE	0.2	–1.3 to 0.7	0.5	–3.2 to 1.0	0.8	–0.1 to 1.0	0.2	–1.3 to 0.7	0.3	–1.3 to 0.8	0.3	–1.0 to 0.8
		AFP	0.03	–	0.08	–	0.2	–	0.03	–	0.06	–	0.06	–
Sleep	Pittsburgh Sleep Quality Index[30] (≥5) (n=102)	OR	2.8	1.2 to 6.9	1.3	0.3 to 7.1	1.0	0.4 to 3.1	**2.8**	**1.2 to 6.9**	1.3	0.5 to 3.5	2.1	0.9 to 5.3
		AFE	0.6	0.2 to 0.9	0.3	–2.7 to 0.9	0.05	–1.9 to 0.7	0.6	0.2 to 0.9	0.2	–1.1 to 0.7	0.5	–0.2 to 0.8
		AFP	0.4	–	0.1	–	0.02	–	0.4	–	0.1	–	0.3	–
	Epworth Sleepiness Scale[31] (>10) (n=105)	OR	1.6	0.6 to 4.8	4.6	0.9 to 22	1.7	0.5 to 5.6	1.6	0.6 to 4.8	1.4	0.4 to 4.2	2.1	0.7 to 6.0
		AFE	0.4	–0.7 to 0.8	0.8	–0.08 to 1.0	0.4	–1.2 to 0.8	0.4	– 0.7 to 0.8	0.3	–1.4 to 0.8	0.2	–0.4 to 0.8
		AFP	0.1	–	0.4	–	0.1	–	0.1	–	0.07	–	0.1	–
Nutrition	Low Energy Availability in Females Questionnaire (≥8)[15]	OR	**3.8**	**1.3 to 12**	1.7	0.3 to 10	**5.8**	**1.6 to 24**	**3.8**	**1.3 to 12**	**4.4**	**1.4 to 14**	1.3	0.4 to 3.5
		AFE	0.7	0.2 to 0.9	0.4	–2.4 to 0.9	0.8	0.4 to 1.0	0.7	0.2 to 0.9	0.8	0.3 to 0.9	0.2	–1.2 to 0.7
		AFP	0.4	–	0.2	–	0.6	–	0.4	–	0.5	–	0.09	–

Note: eye and other illness categories not reported due to the low prevalence.
OR and CI unable to be calculated due to zero responses under the exposure in the control group (zero cells could, note: CIs are derived from Cornfield in this case).
CIs excluding 1.0 are represented in bold to indicate statistical significance.
AFE, attributable fractions in the exposed; AFP, attributable fractions in the population; Any symptoms, reporting any symptoms across all illness categories; Chest, symptoms of chest infection; Est, point estimate; Fatigue, symptoms of fatigue; GIT, gastrointestinal symptoms; Head, symptoms relating to the head region; Injury, sustaining an injury; n, represents the responses to both the questionnaire of interest and the health background questionnaire; Statistic, statistical method; URT, upper respiratory tract infection symptoms.

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Table 5 Pooled and stratified factor analysis of aspects of health

Population	Variable	Factor analysis		Cumulative proportion
		Factor 1	Factor 2	
Pooled factors independent of health status	Age	–0.36	–0.25	
	Depression	**0.71**	–0.20	
	Anxiety	**0.72**	0.22	
	Stress	**0.75**	–0.13	
	Dispositional resilience	–0.28	**0.55**	
	LEAF-Q	0.29	0.15	
	Overall stress	0.15	**0.64**	
	Overall recovery	0.07	**0.77**	
	Epworth	0.20	–0.12	
	Pittsburgh	**0.54**	–0.09	
	Travel	0.09	0.29	0.74
Illness-related factors (sports incapacity)	Age	–0.43	–0.12	
	Depression	0.34	**0.61**	
	Anxiety	**0.74**	0.31	
	Stress	0.40	**0.62**	
	Dispositional resilience	0.18	–0.46	
	LEAF-Q	0.24	0.10	
	Overall stress	**0.78**	–0.23	
	Overall recovery	**0.75**	**–0.50**	
	Sport-related stress	**0.54**	–0.25	
	Sport-related recovery	0.37	–0.37	
	Epworth	0.07	0.05	
	Pittsburgh	0.29	0.49	
	Travel	0.24	<0.00	0.82

Bold represent significant components of the factor; depression, anxiety and stress as measured by the DASS-21; dispositional resilience as measured by the DRS; overall stress and overall recovery as measured by the Recovery-Stress Questionnaire (52-item); Epworth, Epworth Daytime Sleepiness Scale; Pittsburgh, Pittsburgh Sleep Quality Index; Travel, the number of travel occasions (any location) in the last month (any mode, any country). A negative factor loading indicates a negative direction of the relationship, for example, an increase in age was associated with a lower rate of illness.
DASS-21, Depression, Anxiety and Stress Questionnaire; LEAF-Q, Low Energy Availability in Females Questionnaire.

prevalence of depression, anxiety and stress were similar at both the 9-month and 3-month time points of the Olympic preparation; however, the illness categories were subgrouped in the present study, whereas the previous study reported a single aggregated measure of illness. This variation in classification reduces the power of the analyses. Interestingly, the association between illness and poor mental health did not approach statistical or clinical significance in this study with small point estimates and wide CIs including 1.0 (table 4). This outcome implies that any increase in statistical power is unlikely to impact on this finding. Second, this study profiled the actual Olympic cohort, whereas the previous study[9] used Olympic-eligible athletes, that is, shadow squad members prior to Olympic selections in Australia. Survival bias may exist in this study whereby those athletes who had illness and poor mental health did not make it through selection processes.

Poor sleep quality and daytime sleepiness were highly prevalent but not associated with illness

Poor sleep quality, as assessed by the PSQI, was associated with gastrointestinal tract symptoms in the previous month. Sleep quality was not associated with any other illness category and neither was daytime sleepiness, as measured by the Epworth Daytime Sleepiness Scale. These outcomes are interesting given the evidence linking sleep with overall health.[19] Two explanations are offered. First, the most plausible explanation is that GIT symptoms had interrupted sleep and therefore affect sleep quality. Second, the association between GIT symptoms and poor sleep quality may either represent a type 1 error given the lack of other associations in other domains. Furthermore, given the analysis employed compared groups rather than individual, an ecological fallacy may be present. Ecological fallacy is a limitation of cross-sectional studies. This result is therefore reported with caution.

Independent of the association with illness, the prevalence of poor sleep quality was high, with one in two athletes above the clinical threshold (scoring higher than 5 on the PSQI). This outcome confirms the earlier study of a similar cohort[9 20] with up to 83% of participants above the threshold for poor sleep. In the first study, the cohort was recruited around Olympic selections, whereas in the present study, the cohort were recruited following Olympic selections and completing final preparations before the Olympic Games. The lower prevalence (measured 3–4 months later) later might reflect lower psychological and physical pressures after obtaining Olympic selection. Again, a survivor bias might be exerting some effect in relation to mental health.

Lack of risk factor identification

The lack of clear risk factors for illness elicits two plausible explanations: either the chosen factors in this study did not appropriately identify those athletes at risk or illness (and host defence) is a complex system that cannot easily be divided into discrete factors. Other studies have highlighted the difficulties of screening for risk factors for injury in sport.[14] It is likely that many of these difficulties are also the case in relation to illness. One clear difference between illness and injury, however, is injuries are considered non-communicable. This characteristic is in contrast to illness whereby some of the conditions can be contracted though communicable disease pathways.[21] A fundamental of infectious disease control is that susceptibility is not universal.[22] In prevention of sport-related illnesses, a detailed understanding of why some athletes are susceptible and others are not is a key question for researchers and practitioners. This approach may offer solutions beyond simple identification of risk factors and their control to include 'salutogenesis' in which the factors that support good health are prioritised.[23] These salutogenic factors can then be developed, introduced and monitored in high-performance sport programmes. Given the poor hygiene practices of a minority of athletes,[9] heightened vigilance[21] may be warranted in these situations where travel can increase environmental exposure and athletes often cohabitate for accommodation.

Limitations

This research was cross-sectional and analysed as a case–control. As such, no causal inference is provided. There were no healthy controls available within our sample frame. This outcome can be considered a limitation yet also represents the real-world scenario whereby athletes are ill frequently. The use of controls in this manner may underestimate the true effect of the results, and replication of the study is warranted. Data analysed were based on athlete self-reports and may be affected by recall bias and lower health literacy with the corresponding overestimation

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or underestimation of some aspects. This study used valid and reliable questions to ensure these risks were mitigated; however, they cannot be removed entirely. This study does not report p values to limit the number of type 1 error when undertaking multiple analyses. To overcome this issue, all data are presented with CIS such that the reader can ascertain the level of uncertainty in estimated effects in relation to clinically important thresholds.[24]

All participants in this study reported illness symptoms in the previous month. It is likely that this observation indicates some form of recruitment bias; however, all athletes were recruited through their National Sporting Organisation, which encouraged participation. The total participants in this study was 206; however, the health data were absent in 74 participants. The participants who did not report health data may represent a healthy cohort who did not see value in completing the full protocol. While acknowledging the limitations of this study, the outcomes highlight the need for vigilance and attention throughout the preparatory period by every athlete.

CONCLUSION

High prevalence of illness, particularly related to the upper respiratory tract, was reported in Olympic-level athletes in the lead up to the 2016 Olympic Games. LEA in females, as measured by the LEAF-Q, was associated with increased likelihood of upper respiratory tract, gastrointestinal tract, bodily aches and head-related symptoms in the previous month. Components of health are not independent and are highly inter-related, underlining the priority for multidisciplinary prevention and management programmes in high-level sports.

What are the findings?

- Low energy availability in female athletes was highly associated with illness. Energy availability should be monitored via indirect or direct means.
- Most illnesses last 2–7 days; however, high variability in these estimates is reported. Individualised care is recommended.
- There was a high prevalence of sleep disturbances, poor mental health symptoms and illness (primarily related to the upper respiratory tract) in Olympic athletes.

How might it impact on clinical practice in the future?

- Illness prevention programme should include evaluation of the effect of multifactorial preventive measures.
- Good mental health, dispositional resilience and sleep quality appear to be leading factors involved in remaining healthy and should form part of a salutogenic programme.
- Illness prevention is required in the months prior to the Olympic Games.

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Contributors All authors contributed to the design of the study. MD, NV and NPW were responsible for the data collection with input by all remaining researchers. MD and MW were responsible for the data analysis and reporting. All authors provided input to the drafting and final approval of this manuscript.

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Data sharing statement The authors are willing to discuss data sharing under collaborative agreements. Please contact the corresponding author.

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REFERENCES

1 Bittencourt NF, Meeuwisse WH, Mendonça LD, *et al*. Complex systems approach for sports injuries: moving from risk factor identification to injury pattern recognition-narrative review and new concept. *Br J Sports Med* 2016;50:1309–14.
2 Mooney M, Charlton PC, Soltanzadeh S, *et al*. Who 'owns' the injury or illness? Who 'owns' performance? Implications of applying systems thinking to integrate health and performance in elite sport. *Br J Sports Med* 2017:1–2.
3 Drew MK, Raysmith BP, Charlton PC. Injuries impair the chance of success in athletes: a systematic review. *Br J Sports Med* (Epub ahead of print 26 Apr 2017).
4 Gallagher J, Needleman I, Ashley P, *et al*. Self-reported outcome measures of the impact of injury and illness on athlete performance: a systematic review. *Sports Med* 2017;47:1335–48.
5 Schwellnus M, Soligard T, Alonso JM, *et al*. How much is too much? (Part 2) International Olympic Committee consensus statement on load in sport and risk of illness. *Br J Sports Med* 2016;50:1043–52.
6 Soligard T, Steffen K, Palmer D, *et al*. Sports injury and illness incidence in the Rio de Janeiro 2016 Olympic Summer Games: A prospective study of 11274 athletes from 207 countries. *Br J Sports Med* 2017;51:1265–71.
7 Raysmith BP, Drew MK. Performance success or failure is influenced by weeks lost to injury and illness in elite Australian track and field athletes: A 5-year prospective study. *J Sci Med Sport* 2016;19:778–83.
8 Svendsen IS, Gleeson M, Haugen TA, *et al*. Effect of an intense period of competition on race performance and self-reported illness in elite cross-country skiers. *Scand J Med Sci Sports* 2015;25:846–53.
9 Drew MK, Vlahovich N, Hughes D, *et al*. A multifactorial evaluation of illness risk factors in athletes preparing for the Summer Olympic Games. *J Sci Med Sport* 2017;20:745–50.
10 Timpka T, Jacobsson J, Ekberg J, *et al*. Meta-narrative analysis of sports injury reporting practices based on the Injury Definitions Concept Framework (IDCF): a review of consensus statements and epidemiological studies in athletics (track and field). *J Sci Med Sport* 2015;18:643–50.
11 Foster C, Florhaug JA, Franklin J, *et al*. A new approach to monitoring exercise training. *J Strength Cond Res* 2001;15:109–15.
12 Miettinen OS. Proportion of disease caused or prevented by a given exposure, trait or intervention. *Am J Epidemiol* 1974;99:325–32.
13 de Winter JC, Dodou D, Wieringa PA. Exploratory Factor Analysis With Small Sample Sizes. *Multivariate Behav Res* 2009;44:147–81.
14 Bahr R. Why screening tests to predict injury do not work-and probably never will: a critical review. *Br J Sports Med* 2016;50:776–80.
15 Melin A, Tornberg AB, Skouby S, *et al*. The LEAF questionnaire: a screening tool for the identification of female athletes at risk for the female athlete triad. *Br J Sports Med* 2014;48:540–5.
16 Woods AL, Garvican-Lewis LA, Lundy B, *et al*. New approaches to determine fatigue in elite athletes during intensified training: resting metabolic rate and pacing profile. *PLoS One* 2017;12:e0173807.
17 Woods AL, Garvican-Lewis LA, Rice A, *et al*. 12 days of altitude exposure at 1800 m does not increase resting metabolic rate in elite rowers. *Appl Physiol Nutr Metab* 2017;42:672–6.
18 Woods AL, Sharma AP, Garvican-Lewis LA, *et al*. Four Weeks of Classical Altitude Training Increases Resting Metabolic Rate in Highly Trained Middle-Distance Runners. *Int J Sport Nutr Exerc Metab* 2017;27:83–90.
19 Grandner MA. Sleep, Health, and Society. *Sleep Med Clin* 2017;12:1–22.
20 Swinbourne R, Gill N, Vaile J, *et al*. Prevalence of poor sleep quality, sleepiness and obstructive sleep apnoea risk factors in athletes. *Eur J Sport Sci* 2016;16:850–8.
21 Mast EE, Goodman RA. Prevention of infectious disease transmission in sports. *Sports Med* 1997;24:1–7.

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22 Giesecke J. Modern infectious disease epidemiology: Arnold. *Hodder Headline Group* 2001.
23 Lindstrom B. Salutogenesis. *J Epi Comm Health* 2005;59:440–2.
24 Rothman KJ. Curbing type I and type II errors. *Eur J Epidemiol* 2010;25:223–4.
25 Henry JD, Crawford JR. The short-form version of the Depression Anxiety Stress Scales (DASS-21): construct validity and normative data in a large non-clinical sample. *Br J Clin Psychol* 2005;44:227–39.
26 Antony MM, Bieling PJ, Cox BJ, *et al*. Psychometric properties of the 42-item and 21-item versions of the Depression Anxiety Stress Scales in clinical groups and a community sample. *Psychol Assess* 1998;10:176–81.
27 Funk SC. Hardiness: a review of theory and research. *Health Psychol* 1992;11:335–45.
28 Kellmann M, Kallus KW.Recovery-stress questionnaire for athletes:user manual: human kinetics. 2001.
29 Cohen S, Kamarck T, Mermelstein R. A global measure of perceived stress. *J Health Soc Behav* 1983;24:385–96.
30 Buysse DJ, Reynolds CF, Monk TH, *et al*. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and research. *Psychiatry Res* 1989;28:193–213.
31 Johns MW. A new method for measuring daytime sleepiness: the Epworth sleepiness scale. *Sleep* 1991;14:540–5.

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Prevalence of illness, poor mental health and sleep quality and low energy availability prior to the 2016 Summer Olympic Games

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